

**Version: 1.0**

# Technical Specification

MODEL NO: SA1452-EHA  
( EC058TC1 )

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Please contact E Ink or its agent for further information.

☐ Customer's Confirmation

Customer \_\_\_\_\_

Date \_\_\_\_\_

By \_\_\_\_\_

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## Revision History

| Rev. | Issued Date    | Revised / Contents |
|------|----------------|--------------------|
| 1.0  | Jun, 23 , 2020 | First initial      |
|      |                |                    |

# ***TECHNICAL SPECIFICATION***

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## 1. General Description

SA1452-EHA (EC058TC1) as a reflective electrophoretic E Ink® technology display module with color based on active matrix TFT substrate. It has 5.84" active area with 720 x 1440 pixels, the display is capable to display images with Black/White at 2-16 gray levels (1-4 bits), color mode at 480 x 240 pixels with 4096 color depending on the display controller and the associated waveform file it used.

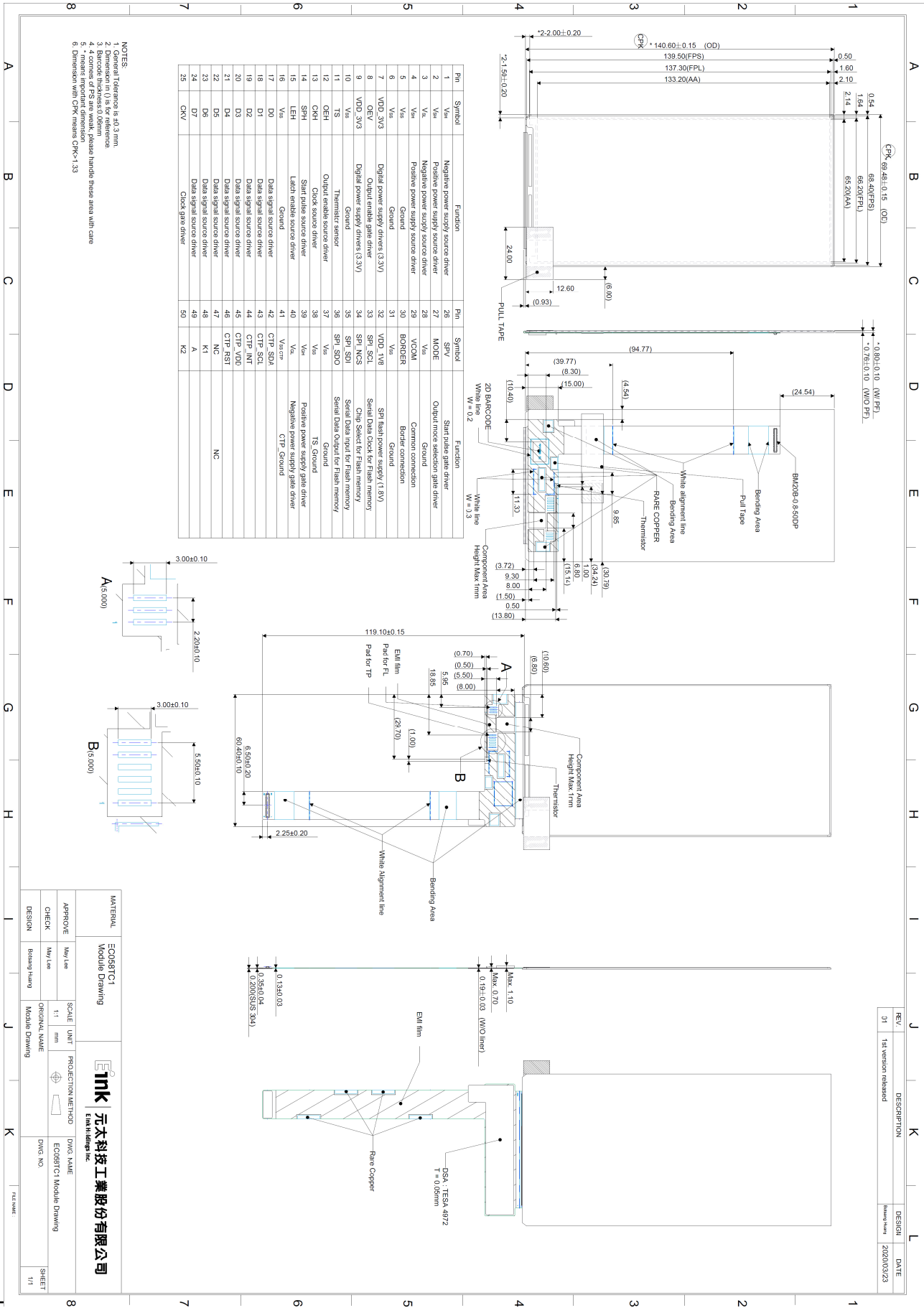
## 2. Features

- Carta High contrast reflective/electrophoretic technology
- 4096 colour
- 720 x 1440 B/W display, 240 x 480 colour display
- High reflectance
- Ultra wide viewing angle
- Ultra low power consumption
- Pure reflective mode
- Bi-stable
- Commercial temperature range
- Portrait mode

## 3. Mechanical Specifications

| Parameter              | Specifications                           | Unit  | Remark     |
|------------------------|------------------------------------------|-------|------------|
| Screen Size            | 5.84                                     | Inch  |            |
| Display Resolution     | B/W 720 (H)×1440(V), Color 240(H)×480(V) | Pixel | 4096 color |
| Active Area            | 65.16 (H)×133.2 (V)                      | mm    |            |
| Pixel Pitch            | 90.5 (H) × 92.5 (V)                      | μm    |            |
| Pixel Configuration    | Rectangle                                |       |            |
| Outline Dimension      | 69.48 (W) × 140.60 (H) × 0.76 (D)        | mm    | w/o PF     |
| Module Weight          | 15.9+/-1.6                               | g     | w/o PF     |
| Number of Gray         | 16 Gray Level (monochrome)               |       |            |
| Display operating mode | Reflective mode                          |       |            |

## 4. Mechanical Drawing of Display Module



## 5. Input/output Interface

5-1) Connector type: BM20B(0.8)-50DP-0.4V(51)-ND

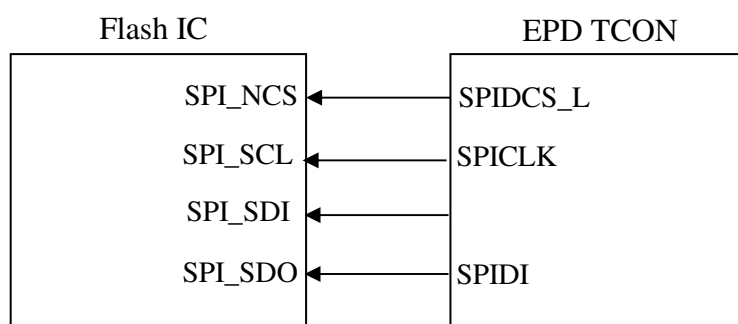
5-2) Pin Assignment

| Pin # | Signal             | I/O | Description                         | Remark |
|-------|--------------------|-----|-------------------------------------|--------|
| 1     | V <sub>SL</sub>    | P   | Negative power supply source        |        |
| 2     | V <sub>SH</sub>    | P   | Positive power supply source        |        |
| 3     | V <sub>SL</sub>    | P   | Negative power supply source        |        |
| 4     | V <sub>SH</sub>    | P   | Positive power supply source        |        |
| 5     | V <sub>SS</sub>    | P   | Ground                              |        |
| 6     | V <sub>SS</sub>    | P   | Ground                              |        |
| 7     | V <sub>DD</sub>    | P   | Digital power supply drivers (3.3V) |        |
| 8     | OE <sub>V</sub>    | I   | Output enable of gate driver        |        |
| 9     | V <sub>DD</sub>    | P   | Digital power supply drivers (3.3V) |        |
| 10    | V <sub>SS</sub>    | P   | Ground                              |        |
| 11    | TS                 | -   | Thermistor sensor                   |        |
| 12    | OE <sub>H</sub>    | I   | Output enable source driver         |        |
| 13    | CK <sub>H</sub>    | I   | Clock of source driver              |        |
| 14    | SP <sub>H</sub>    | I   | Start pulse of source driver        |        |
| 15    | LE <sub>H</sub>    | I   | Latch enable of source driver       |        |
| 16    | V <sub>SS</sub>    | P   | Ground                              |        |
| 17    | D <sub>0</sub>     | I   | Data signal source driver           |        |
| 18    | D <sub>1</sub>     | I   | Data signal source driver           |        |
| 19    | D <sub>2</sub>     | I   | Data signal source driver           |        |
| 20    | D <sub>3</sub>     | I   | Data signal source driver           |        |
| 21    | D <sub>4</sub>     | I   | Data signal source driver           |        |
| 22    | D <sub>5</sub>     | I   | Data signal source driver           |        |
| 23    | D <sub>6</sub>     | I   | Data signal source driver           |        |
| 24    | D <sub>7</sub>     | I   | Data signal source driver           |        |
| 25    | CK <sub>V</sub>    | I   | Clock of gate driver                |        |
| 26    | SP <sub>V</sub>    | I   | Start pulse of gate driver          |        |
| 27    | MODE               | I   | Output mode selection gate driver   |        |
| 28    | V <sub>SS</sub>    | P   | Ground                              |        |
| 29    | V <sub>COM</sub>   | P   | Common connection                   |        |
| 30    | Border             | P   | Border connection                   |        |
| 31    | V <sub>SS</sub>    | P   | Ground                              |        |
| 32    | V <sub>DD1.8</sub> | P   | SPI flash power supply (1.8V)       |        |
| 33    | SPI SCL            | I   | Serial Data Clock for Flash         |        |
| 34    | SPI NCS            | I   | Chip Select for Flash memory        |        |
| 35    | SPI SDI            | I   | Serial Data Input for Flash         |        |

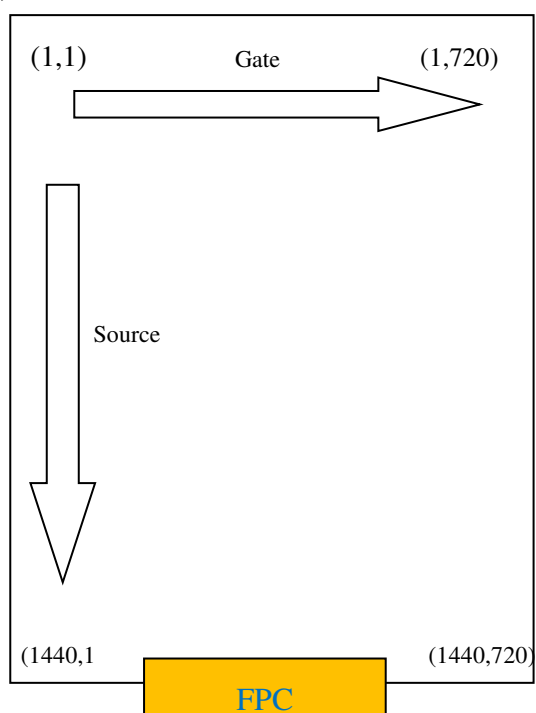
## SA1452-EHA (EC058TC1)

|    |                     |   |                                   |  |
|----|---------------------|---|-----------------------------------|--|
| 36 | SPI SDO             | I | Serial Data Output for Flash      |  |
| 37 | V <sub>SS</sub>     | P | Ground                            |  |
| 38 | V <sub>SS</sub>     | P | Ground                            |  |
| 39 | V <sub>GH</sub>     | P | Positive power supply gate driver |  |
| 40 | V <sub>GL</sub>     | P | Negative power supply gate driver |  |
| 41 | V <sub>SS_CTP</sub> | P | CTP Ground                        |  |
| 42 | CTP SDA             | I | Touch signal                      |  |
| 43 | CTP SCL             | I | Touch signal                      |  |
| 44 | CTP INT             | I | Touch signal                      |  |
| 45 | CTP VDD             | I | Touch signal                      |  |
| 46 | CTP RST             | I | Touch signal                      |  |
| 47 | NC                  | - | NO Connection                     |  |
| 48 | K1                  | - | Front light signal                |  |
| 49 | A                   | - | Front light signal                |  |
| 50 | K2                  | - | Front light signal                |  |

### Note 5-1



### 5-3) Panel Scan Direction



## 6. Electrical Characteristics

### 6-1) Absolute Maximum Ratings:

| Parameter                | Symbol                            | Rating      | Unit | Remark |
|--------------------------|-----------------------------------|-------------|------|--------|
| Logic Supply Voltage     | VDD_3V3                           | -0.3 to 5.0 | V    | --     |
| Positive Supply Voltage  | V <sub>SH</sub>                   | -0.3 to +18 | V    | --     |
| Negative Supply Voltage  | V <sub>SL</sub>                   | -18 to +0.3 | V    | --     |
| Max .Drive Voltage Range | V <sub>SH</sub> – V <sub>SL</sub> | 36          | V    | --     |
| Supply Voltage           | V <sub>GH</sub>                   | -0.3 to 46  | V    | --     |
| Supply Voltage           | V <sub>GL</sub>                   | -25 to +0.3 | V    | --     |
| Supply Range             | V <sub>GH</sub> -V <sub>GL</sub>  | -0.3 to +46 | V    | --     |
| Operating Temp. Range    | TOTR                              | 0 to +50    | °C   | --     |
| Storage Temperature      | TSTG                              | -25 to 70   | °C   | --     |

### 6-2) Panel DC Characteristics

| Parameter              | Symbol               | Conditions                        | Min   | Typ      | Max     | Unit |
|------------------------|----------------------|-----------------------------------|-------|----------|---------|------|
| Signal ground          | V <sub>SS</sub>      |                                   | -     | 0        | -       | V    |
| Logic Voltage supply   | VDD_3V3              |                                   | 3.0   | 3.3      | 3.6     | V    |
|                        | I <sub>VDD_3V3</sub> | V <sub>DD</sub> =3.3V             | -     | 9.42     | 10.01   | mA   |
| SPI Voltage supply     | VDD_1V8              |                                   | 1.65  | 1.8      | 1.95    | V    |
|                        | I <sub>VDD_1V8</sub> | V <sub>DD_1V8</sub> =1.8V         | -     |          | 1.5     | mA   |
| Gate Negative supply   | V <sub>GL</sub>      |                                   | -22   | -21      | -20     | V    |
|                        | I <sub>GL</sub>      | V <sub>GL</sub> = -20V            | -     | 3.55     | 11.62   | mA   |
| Gate Positive supply   | V <sub>GH</sub>      |                                   | 21    | 22       | 23      | V    |
|                        | I <sub>GH</sub>      | V <sub>GH</sub> = 22V             | -     | 0.79     | 2.28    | mA   |
| Source Negative supply | V <sub>SL</sub>      |                                   | -15.4 | -15      | -14.6   | V    |
|                        | I <sub>SL</sub>      | V <sub>SL</sub> = -15V            | -     | 35       | 87.02   | mA   |
| Source Positive supply | V <sub>SH</sub>      |                                   | 14.6  | 15       | 15.4    | V    |
|                        | I <sub>SH</sub>      | V <sub>SH</sub> = 15V             | -     | 39.54    | 100.37  | mA   |
| Border supply          | V <sub>COM</sub>     |                                   | -     | Adjusted | -       | V    |
| Asymmetry source       | V <sub>Asym</sub>    | V <sub>SH</sub> + V <sub>SL</sub> | -800  | 0        | 800     | mV   |
| Common voltage         | V <sub>COM</sub>     |                                   | -4    | Adjusted | -0.5    | V    |
|                        | I <sub>COM</sub>     |                                   | -     | 1.33     | 4.18    | mA   |
| Total Power            | P                    |                                   | -     | 1240.81  | 3163.24 | mW   |
| Standby power          | P <sub>STBY</sub>    |                                   | -     | -        | 0.4     | mW   |

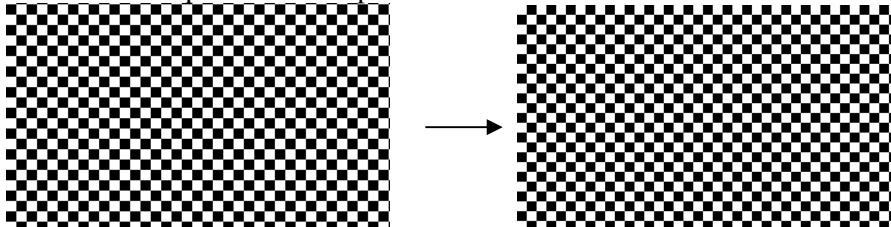


| Parameter        | Symbol           | Conditions             | Min | Typ | Max | Unit |
|------------------|------------------|------------------------|-----|-----|-----|------|
| Maximum Currents | I <sub>GL</sub>  | V <sub>GL</sub> = -20V | -   | -   | 158 | mA   |
|                  | I <sub>GH</sub>  | V <sub>GH</sub> = +22V | -   | -   | 22  | mA   |
|                  | I <sub>SL</sub>  | V <sub>SL</sub> = -15V | -   | -   | 493 | mA   |
|                  | I <sub>SH</sub>  | V <sub>SH</sub> = +15V | -   | -   | 549 | mA   |
|                  | I <sub>COM</sub> | -                      | -   | -   | 272 | mA   |

- The maximum power consumption is measured using 85 Hz waveform with following pattern transition: from pattern of repeated 1 consecutive black scan lines followed by 1 consecutive white scan line to that of repeated 1 consecutive white scan lines followed by 1 consecutive black scan lines. (Note 6-1)
- The Typical power consumption is measured using 85 Hz waveform with following pattern transition: from horizontal 4 gray scale pattern to vertical 4 gray scale pattern. (Note 6-2)
- The standby power is the consumed power when the panel controller is in standby mode.
- The listed electrical/optical characteristics are only guaranteed under the controller & waveform provided by E Ink.
- V<sub>com</sub> is recommended to be set in the range of assigned value  $\pm 0.1$  V

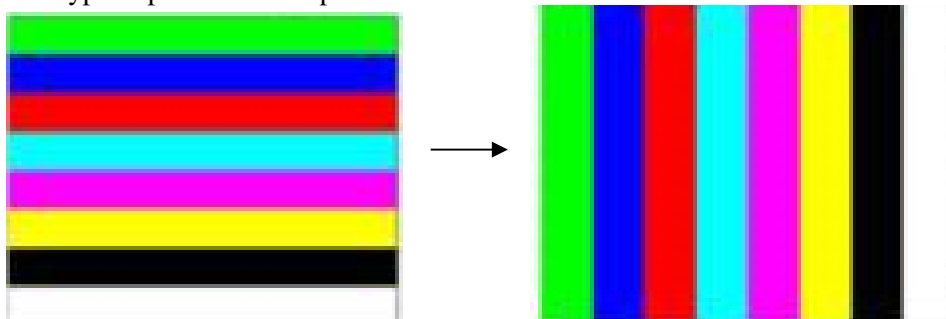
Note6-1

The maximum power consumption



Note 6-2

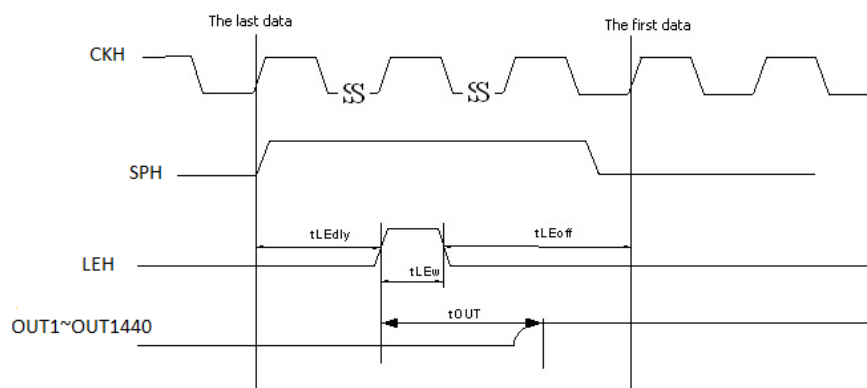
The typical power consumption



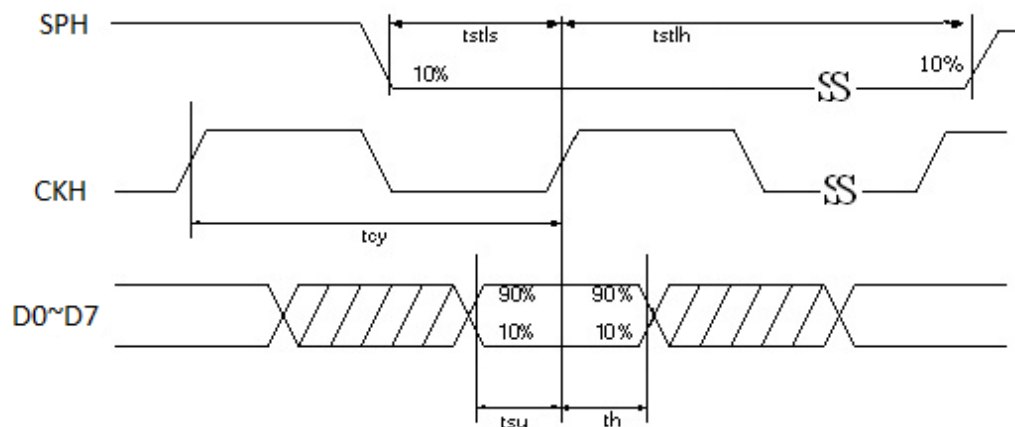
#### 6-4) Panel AC characteristics

| Parameter                                                    | Symbol | Min.    | Typ. | Max.          | Unit |
|--------------------------------------------------------------|--------|---------|------|---------------|------|
| Clock frequency                                              | fckv   |         |      | 200           | kHz  |
| Minimum “L” clock pulse width                                | twL    | 0.5     |      |               | us   |
| Minimum “H” clock pulse width                                | twH    | 0.5     |      |               | us   |
| Clock rise time                                              | trckv  |         |      | 100           | ns   |
| Clock fall time                                              | tfckv  |         |      | 100           | ns   |
| SPV setup time                                               | tSU    | 100     |      | twH-100       | ns   |
| SPV hold time                                                | tH     | 100     |      | twH-100       | ns   |
| Pulse rise time                                              | trspv  |         |      | 100           | ns   |
| Pulse fall time                                              | tfspv  |         |      | 100           | ns   |
| Clock CKH cycle time                                         | tcy    | 16.67   | 50   |               | ns   |
| D0 .. D7 setup time                                          | tsu    | 8       |      |               | ns   |
| D0 .. D7 hold time                                           | th     | 8       |      |               | ns   |
| SPH setup time                                               | tstls  | 0.5*tcy |      | 0.8*tcy       | ns   |
| SPH hold time                                                | tstlh  | 0.5*tcy |      | 240*tcy-tstls | ns   |
| LEH on delay time                                            | tLEdly | 3.5*tcy |      |               | ns   |
| LEH high-level pulse width<br>(When VDD=1.7V to 2.1V)        | tLEw   | 300     |      |               | ns   |
| LEH off delay time                                           | tLEoff | 200     |      |               | ns   |
| Output setting time to +/-<br>30mV(C <sub>load</sub> =200pF) | tout   |         |      | 20            | us   |

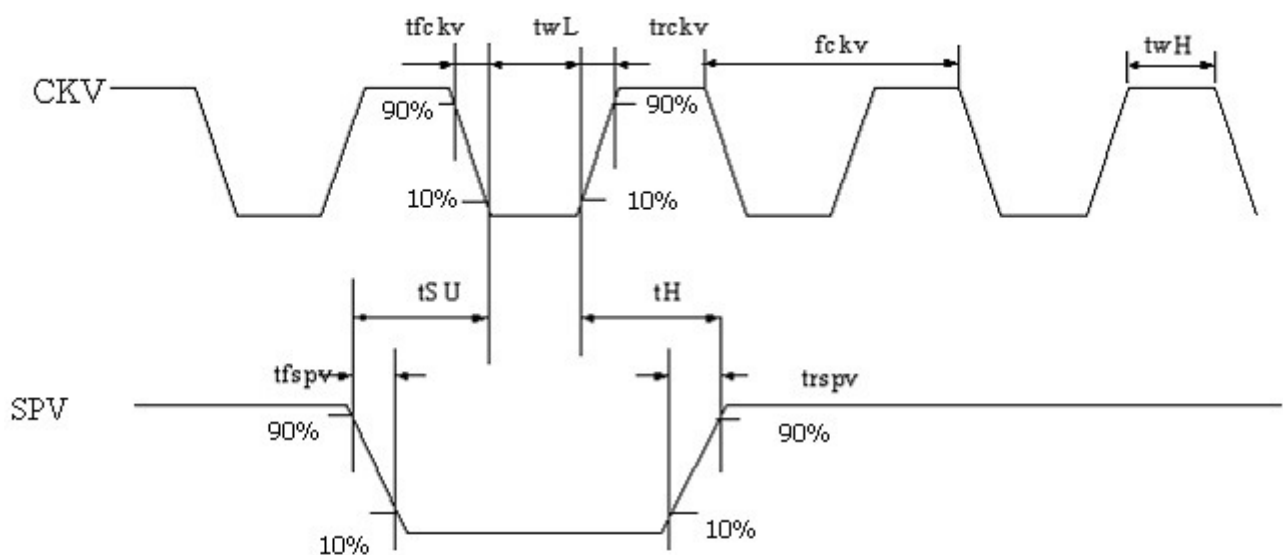
#### OUTPUT LATCH CONTROL SIGNALS



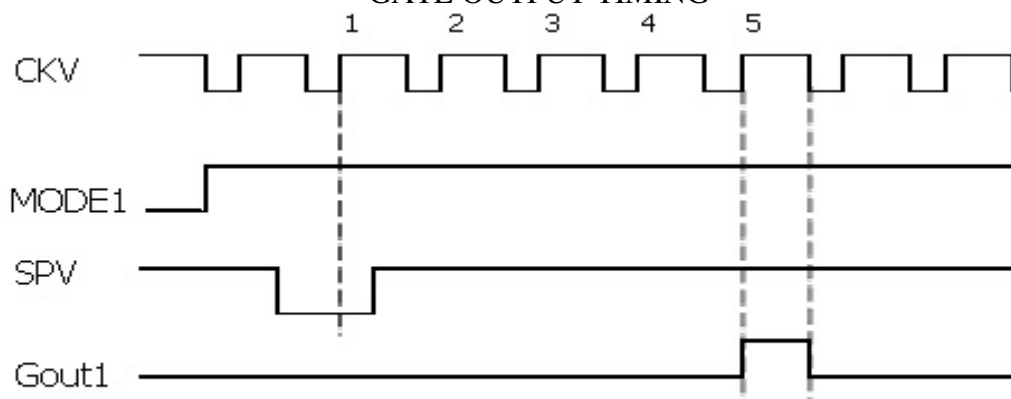
#### CLOCK & DATA TIMING



### CKV & SPV TIMING



### GATE OUTPUT TIMING



**Note :** First gate line on timing after 5CKV gate line is on

### 6-5) Refresh Rate

The module is applied at a maximum refresh rate of 85 Hz.

|                     | Min | Max   |
|---------------------|-----|-------|
| <b>Refresh Rate</b> | -   | 85 Hz |

### 6-6) Controller Timing

This timing mode is depicted on Figure 1 and Figure 2 and it refers to timing of Source Driver Output Enable (OEH) and Gate Driver Clock (CKV). Note, that in this mode LGON follows CKV timing.

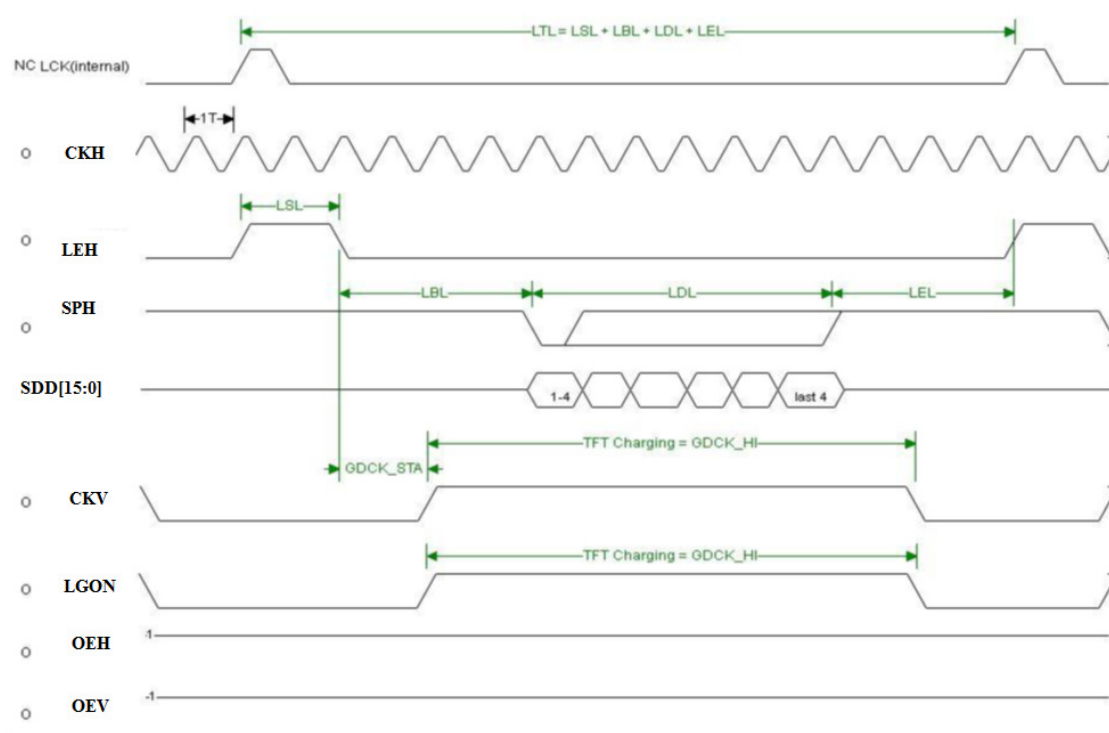


Figure 1. Line Timing in Mode 3

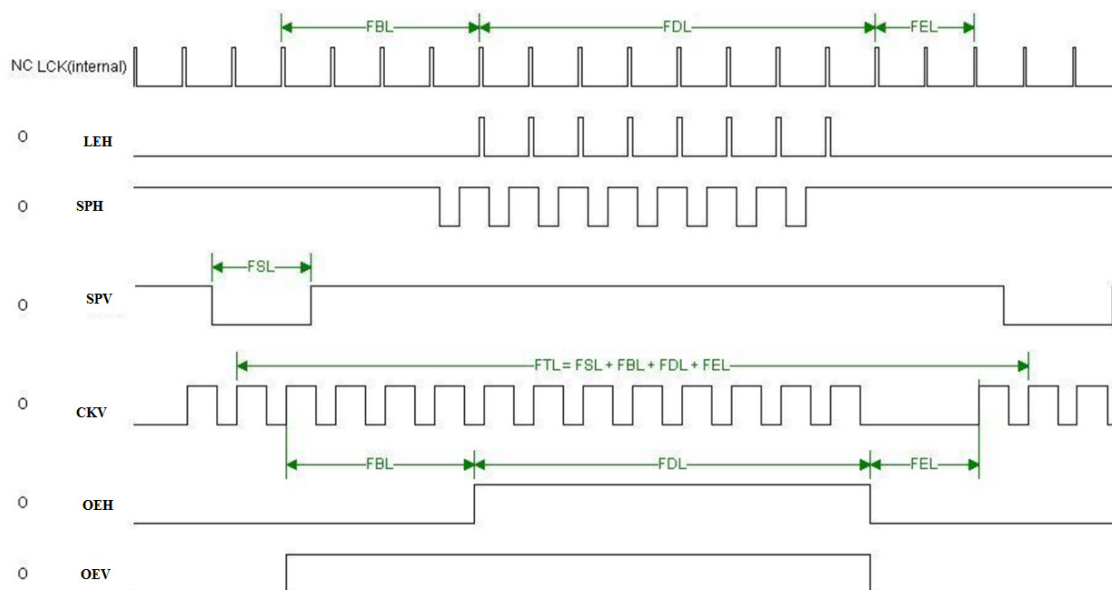


Figure 2 Frame Timing in Mode 3

Timing Parameters Table

| Mode               | 3     | Resolution<br>1440 x 720 |          |        |         |         |
|--------------------|-------|--------------------------|----------|--------|---------|---------|
| CKH [MHz]          | 24    |                          |          |        |         |         |
| Pixels Per CKH     | 4     |                          |          |        |         |         |
| Line               | LSL   | LBL                      | LDL      | LEL    | CKV_STA | LGONL   |
| Parameters[CKH]    | 10    | 6                        | 360      | 6      | 4       | 354     |
| Line               | -     | -                        | -        | -      | -       | -       |
| Parameters[us]     | 0.42  | 0.25                     | 15.00    | 0.25   | 0.17    | 14.75   |
| Frame              | FSL   | FBL                      | FDL      | FEL    | -       | FR [Hz] |
| Parameters [lines] | 1     | 4                        | 720      | 14     | -       | 85.02   |
| Frame              | -     | -                        | -        | -      | -       | -       |
| Parameters [us]    | 15.92 | 63.67                    | 11460.00 | 222.83 | -       | -       |

Note 1: For parameters definition, see Section 7. Active Matrix Electronic Paper Display Timings

Note 2: For Isis Controller CKV\_STA and LGONL are not settable parameters; CKV\_STA = LBL, LGONL = LDL+0.5

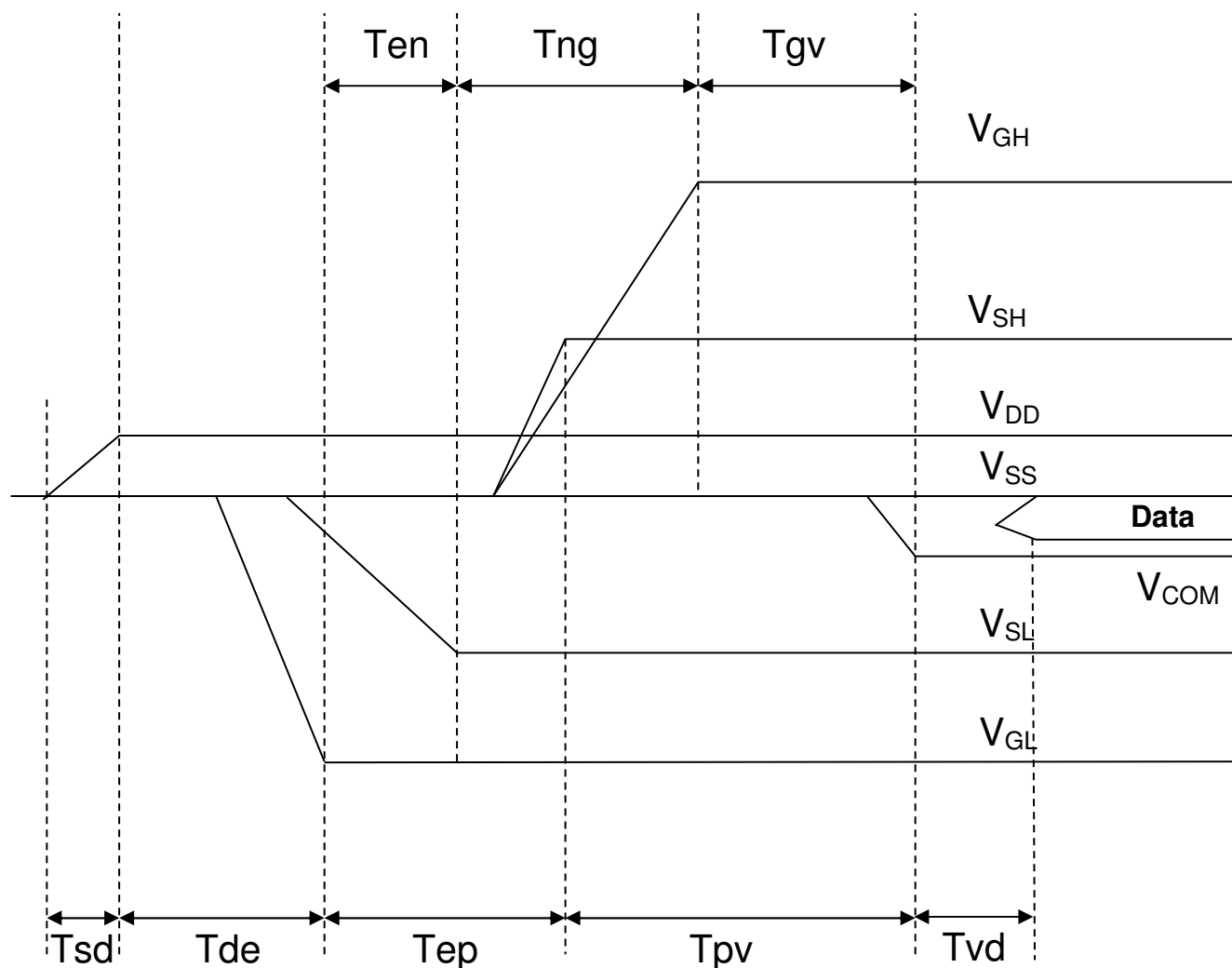
Note 3: For Freescale SoC OEV Low pulse represent FSL and SPV pulses with the first period of FBL

## 7. Power Sequence

Power Rails must be sequenced in the following order :

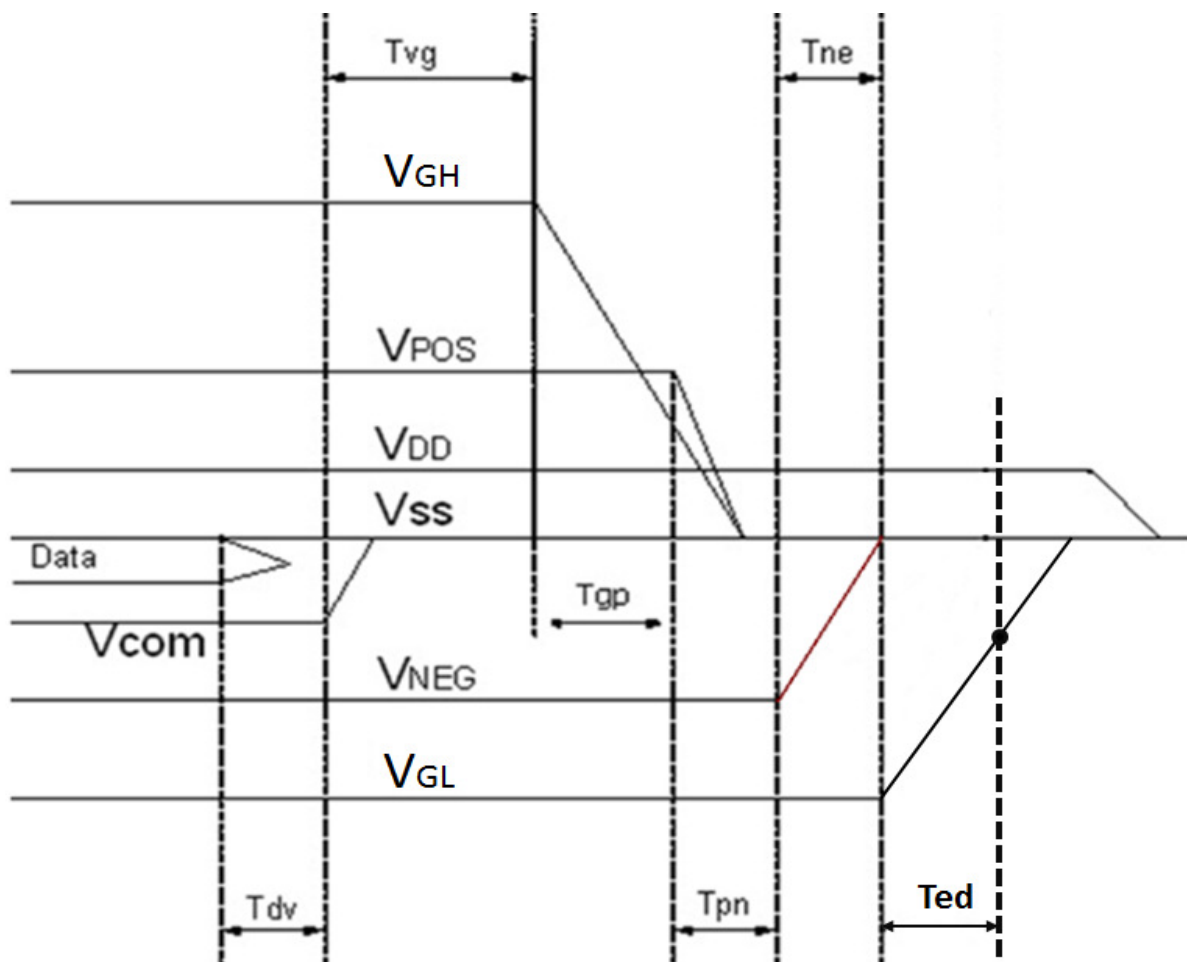
1.  $V_{SS} \rightarrow V_{DD} \rightarrow V_{SL} \rightarrow V_{SH}$  (Source Driver)  $\rightarrow V_{COM}$
2.  $V_{SS} \rightarrow V_{DD} \rightarrow V_{GL} \rightarrow V_{GH}$  (Gate Driver)

### POWER ON



|     | Min    | Max |
|-----|--------|-----|
| Tsd | 30us   | -   |
| Tde | 100us  | -   |
| Tep | 1000us | -   |
| Tpv | 100us  | -   |
| Tvd | 100us  | -   |
| Ten | 0us    | -   |
| Tng | 1000us | -   |
| Tgv | 100us  | -   |

## POWER OFF



|     | Min   | Max | Remark                       |
|-----|-------|-----|------------------------------|
| Tdv | 100μs | -   | -                            |
| Tvg | 0μs   | -   | -                            |
| Tgp | 0μs   | -   | -                            |
| Tpn | 0μs   | -   | -                            |
| Tne | 0μs   | -   | -                            |
| Ted | 0.5s  | -   | Discharged point @ -7.4 Volt |

Note 7-1 : Supply voltages decay through pull-down resistors.

Note 7-2 : Begin to turn off V<sub>GL</sub> power after V<sub>SL</sub> and V<sub>SH</sub> are completely or almost discharged to GND state.

Note 7-3 : V<sub>GL</sub> must remain negative of Vcom during decay period

## 8. Optical characteristic

### 8-1) Specification

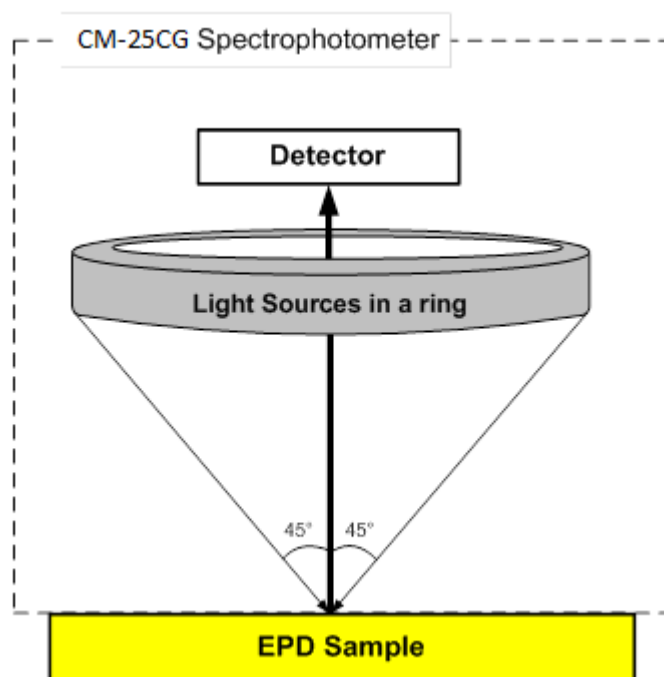
Measurements are made with that the illumination is under an angle of 45 degrees, the detector is perpendicular unless otherwise specified.

T = 25°C

| Symbol      | Parameter                                    | Conditions | Min   | Typ.  | Max   | Unit | Note |
|-------------|----------------------------------------------|------------|-------|-------|-------|------|------|
| R           | Reflectance                                  | White      | 18    | 24    | -     | %    | 8-3  |
| CR          | Contrast Ratio                               | WK         | 10    | 15    | -     |      | 8-4  |
| Color Gamut |                                              | RGBCMYWK   | 1100  | 1570  |       |      | 8-5  |
| Symbol      | Parameter                                    | Conditions | L*TYP | a*TYP | b*TYP | Unit | Note |
| R           | Color coordinates<br>( $\Delta E_{00} < 5$ ) | Red        | 32.81 | 5.29  | 0.12  |      | 8-6  |
| G           |                                              | Green      | 35.99 | -8.52 | 2.97  |      |      |
| B           |                                              | Blue       | 33.04 | -4.19 | -8.16 |      |      |

Note 8-1) W: White , K: Dark , R: Red , G: Green , B: blue , C: Cyan , M: Magenta , Y: Yellow

Note 8-2) Luminance meter: Minolta CM-25cG Spectrophotometer





### 8-3) Reflection Ratio

The reflection ratio is expressed as:

$$R = \text{Reflectance Factor}_{\text{white board}} \times (L_{\text{center}} / L_{\text{white board}})$$

$L_{\text{center}}$  is the luminance measured at center in a white area ( $R=G=B=1$ ).  $L_{\text{white board}}$  is the luminance of a standard white board. Both are measured with equivalent illumination source. The viewing angle shall be no more than 2 degrees.

### 8-4) Definition of contrast ratio

The contrast ratio (CR) is the ratio between the reflectance in a full white area ( $R_l$ ) and the reflectance in a dark area ( $R_d$ ):  $CR = R_l / R_d$

### 8-5 Definition of Color Gamut

Gamut metric: color volume of Red, Green, Blue, Cyan, Magenta, Yellow white, and black in 3D CIE 1976  $L^* a^* b^*$  (CIELAB) color space.

### 8-6) Definition of Color difference ( $\Delta E_{00}$ )

The formula of CIEDE2000 color difference create an uniform color space, which means the  $\Delta E_{00}$  value is equivalent to the perceptual difference in every color.

$$\Delta E_{00} = \left( \left( \frac{\Delta L^*}{k_L S_L} \right)^2 + \left( \frac{\Delta C'}{k_C S_C} \right)^2 + \left( \frac{\Delta H'}{k_H S_H} \right)^2 + R_T \left( \frac{\Delta C' \Delta H'}{S_C S_H} \right) \right)^{1/2}$$

9. Handling, Safety and Environment Requirements and Remark

| <b>WARNING</b>                                                                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The display glass may break when it is dropped or bumped on a hard surface. Handle with care. Should the display break, do not touch the electrophoretic material. In case of contact with electrophoretic material, wash with water and soap. |

| <b>REMARK</b>                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All The specifications listed in this document are guaranteed for module only. Post-assembled operation or component(s) may impact module performance or cause unexpected effect or damage and therefore listed specifications is not warranted after any Post-assembled operation. |

| <b>CAUTION</b>                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------|
| The display module should not be exposed to harmful gases, such as acid and alkali gases, which corrode electronic components. |
| Disassembling the display module can cause permanent damage and invalidate the warranty agreements.                            |
| IPA solvent can only be applied on active area and the back of a glass. For the rest part, it is not allowed.                  |

| <b>Mounting Precautions</b>                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (1) It's recommended that you consider the mounting structure so that uneven force (ex. Twisted stress) is not applied to the module.                                                                                                                                                                                |
| (2) It's recommended that you attach a transparent protective plate to the surface in order to protect the EPD. Transparent protective plate should have sufficient strength in order to resist external force.                                                                                                      |
| (3) You should adopt radiation structure to satisfy the temperature specification.                                                                                                                                                                                                                                   |
| (4) Acetic acid type and chlorine type materials for the cover case are not desirable because the former generates corrosive gas of attacking the PS at high temperature and the latter causes circuit break by electro-chemical reaction.                                                                           |
| (5) Do not touch, push or rub the exposed PS with glass, tweezers or anything harder than HB pencil lead. And please do not rub with dust clothes with chemical treatment. Do not touch the surface of PS for bare hand or greasy cloth. (Some cosmetics deteriorate the PS)                                         |
| (6) When the surface becomes dusty, please wipe gently with absorbent cotton or other soft materials like chamois soaks with petroleum benzene. Normal-hexane is recommended for cleaning the adhesives used to attach the PS. Do not use acetone, toluene and alcohol because they cause chemical damage to the PS. |

(7) Wipe off saliva or water drops as soon as possible. Their long time contact with PS causes deformations and color fading.

**Data sheet status**

Product  
specification

This data sheet contains formal product specifications.

**Limiting values**

Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134). Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or at any other conditions above those given in the Characteristics sections of the specification is not implied. Exposure to limiting values for extended periods may affect device reliability.

**Application information**

Where application information is given, it is advisory and does not form part of the specification.

**10. Reliability test**

|   | TEST                                         | CONDITION                                                         | METHOD           | REMARK |
|---|----------------------------------------------|-------------------------------------------------------------------|------------------|--------|
| 1 | High-Temperature Operation                   | T = +50°C, RH = 30% for 240 hrs                                   | IEC 60 068-2-2Be |        |
| 2 | Low-Temperature Operation                    | T = 0°C for 240 hrs                                               | IEC 60 068-2-1Ae |        |
| 3 | Low-Temperature Storage                      | T = -25°C for 240 hrs<br>Test in white pattern                    | IEC 60 068-2-1Ab |        |
| 4 | High-Temperature,<br>High-Humidity Operation | T = +40°C, RH = 90% for 168 hrs                                   | IEC 60 068-2-78  |        |
| 5 | High Temperature Storage                     | T = +60°C, RH=26% for 240 hrs<br>Test in white pattern            | IEC 60 068-2-78  |        |
| 6 | Temperature Cycle                            | -25°C → +70°C, 100 Cycles<br>30min 30min<br>Test in white pattern | IEC 68-2-14 Nb   |        |
| 7 | Solar radiation test                         | 765 W/m <sup>2</sup> for 168hrs, 40°C<br>Test in white pattern    | IEC60 068-2-5Sa  |        |
| 8 | Electrostatic Effect<br>(non-operating)      | (Machine model)+/- 250V<br>0Ω, 200pF                              | IEC 62180        |        |

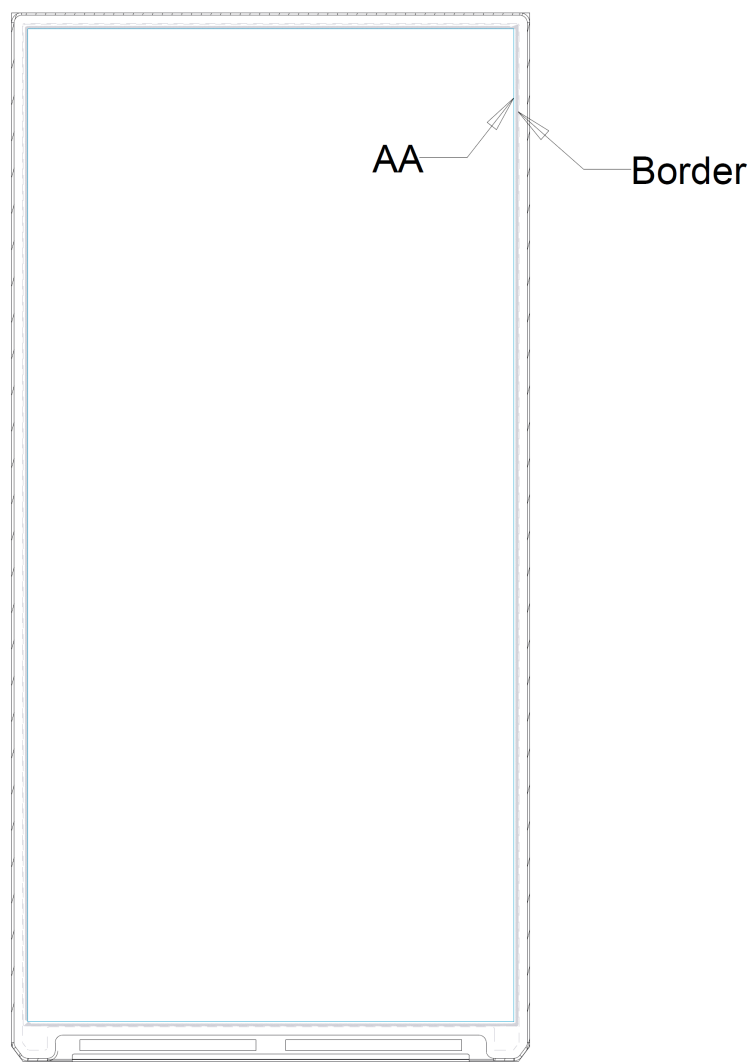
Actual EMC level to be measured on customer application

Note: The protective film must be removed before temperature test.

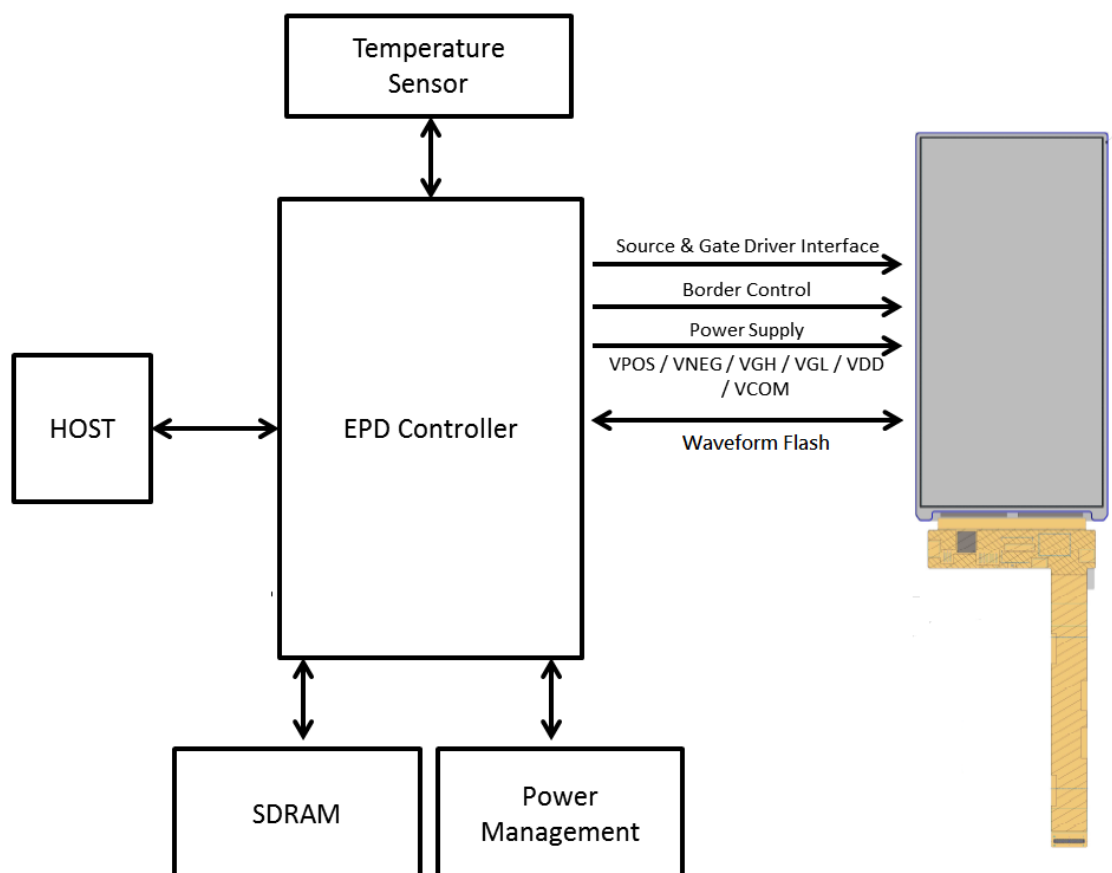
**< Criteria >**

In the standard conditions, there is not display function NG issue occurred. (Including: line defect, no image). All the cosmetic specification is judged before the reliability stress.

11. Border definition



## 12. Block Diagram



### 13. Packing

| REV | DESCRIPTION     | DESIGN        | DATE       |
|-----|-----------------|---------------|------------|
| 01  | INITIAL RELEASE | Botsang Huang | 2020/03/23 |
| 02  |                 |               |            |

**NOTE:**

1. One layer include:  
6 pcs module & 1pcs tray & 1pcs EPE sheet

2. Q'TY: 72 pcs panel/carton.

3. Dimension: 445\*365\*170mm

|      |                       |      |        |
|------|-----------------------|------|--------|
| 7    | 30g加厚複合紙(含乾燥劑)73*95mm | 2    |        |
| 6    | EPE SHEET             | 12   |        |
| 5    | CARTON INTERNAL       | 1    |        |
| 4    | 摺口袋450*380*580mm      | 1    | 抗靜電    |
| 3    | EC058TC1              | 72   |        |
| 2    | TRAY                  | 13   | 抗靜電    |
| 1    | EPE FOAM              | 2    |        |
| ITEM | DESCRIPTION           | Q'TY | REMARK |

| MTL.SPEC. | UNSPECIFIED TOL'S ±5.0mm | REMARK     | <b>Eink</b> 元太科技工業股份有限公司<br>E Ink Holdings Inc. |      |         |                  |
|-----------|--------------------------|------------|-------------------------------------------------|------|---------|------------------|
| APPROVE   | May Lee                  | 2020/03/23 | SCALE                                           | UNIT | SHEET   | DWG.TITLE        |
| CHECK     | May Lee                  | 2020/03/23 | 1:1                                             | mm   | 1 OF 1  | EC058TC1 PACKING |
| DESIGN    | Botsang Huang            | 2020/03/23 | MTL.NO.                                         |      | DWG.NO. | A4<br>SIZE       |